Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 10 · day per timetable	10	1 period (60 min)	Identities with Polynomials	Algebra II	Topic 3 — Polynomial Functions · Lesson 3-3	Grade 10 — 10A and 10C

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Whiteboards · Desmos (checking an identity with two graphs) · Canva (final product) · Squared paper	Lesson presentation (15 slides) · "Choose Your Route" activity (3 pages) · Wayground kick-off

Competencies	Technology Embedded	Connection to student's real life
1- Proving a polynomial identity and disproving a false one with a number. 2- Expanding $(a \pm b)^n$ with Pascal's triangle and the binomial theorem.	1- Quizizz live diagnostic with instant gap analysis. 2- Desmos: graph both sides of an identity and see one curve.	1- A plaza and its walkway: the area is a difference of squares. 2- Checking big products quickly: 99×101.

Values	Soft Skills	Hard Skills
1- Accuracy — the middle term and its sign are written down before they are used. 2- Precision — exact answers first.	1- Collaboration in assigned group roles. 2- Explaining in words why $(a+b)^2$ is not a^2+b^2.	1- Proving an identity by expanding one side. 2- Expanding a binomial power with Pascal's triangle.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Understand and prove polynomial identities. 2- Apply polynomial identities for multiplication, factoring, and simplification. 3- Expand a power of a binomial by applying the binomial theorem. *Vocabulary: Binomial Coefficient ; binomial theorem ; Identity ; Pascal's triangle.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: difference of squares, squaring a product, comparing $(a+b)^2$ with a^2+b^2. Results read as a gap map and used to route students. FIKR Phase 2 — Targeted Instruction. Proving an identity; the square and difference patterns; identities and numbers; Pascal's triangle and the binomial theorem. FIKR Phase 3 — Practice. Two identical "Give it a go" problems; then students choose a route and may switch. Compare routes and judge the claim. Pear Deck feedback. FIKR Phase 4 — Production. Wadi Hanifa plaza and walkway; AI as challenger only. FIKR Phase 5 — Proof of Learning. "Time to Check". No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Make a model identity, prove it, swap and test; publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Whiteboard check (80% threshold) and "Give it a go" — self-marked. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. Final product graded on reasoning and clarity.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell, then: "Which is harder — remembering the middle term, or choosing which identity to use?"
Reflection for Improvement:	
Homework:	Upload the final product to the LMS portfolio. Pear Deck practice set (12 questions, choose your route). Schoology homework quiz for Lesson 3-3. Watch the flipped video for Lesson 3-4. Clinic group: bring your Time to Check paper.
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